Sound: pitch and amplitude

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Learning target: Use a longitudinal-wave model to distinguish pitch and sound level.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 5.*
Away from the normal; frequency stays fixed at a stationary boundary.

02. Worked example: explain the model
Same horizontal spacing of cycles, different vertical excursions.

03. Read and interpret the model
Compressions dense, rarefactions sparse; particles oscillate locally in basic model.

04. Apply the physics
B; 400 and 800 cycles.

05. Explain your reasoning
Amplitude/intensity changes; frequency content need not double.

06. Review: connect an earlier idea *Review: Unit 5, Day 8.*
Work input enables the transfer; it does not contradict the direction of spontaneous heat flow.

07. Exit check
Longitudinal; oscillate parallel.

08. STRETCH: test the model
Pitch change alters frequency/period; amplitude change changes vertical size at constant period.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 5.*
$\theta_c \approx 38.68°$ from the normal. Above it, total internal reflection occurs.

02. Worked example: explain the model
Same horizontal spacing of cycles, different vertical excursions.

03. Read and interpret the model
Compressions dense, rarefactions sparse; particles oscillate locally in basic model.

04. Apply the physics
1.36 m;0.68 m; intensity quadruples.

05. Explain your reasoning
Amplitude/intensity changes; frequency content need not double.

06. Review: connect an earlier idea *Review: Unit 5, Day 8.*
Total energy is conserved. At common temperature the initial temperature-difference opportunity for work is eliminated.

07. Exit check
50 cycles; T=0.002 s.

08. STRETCH: test the model
5 beats/s is amplitude modulation of the combined sound; original tones remain near 400 Hz.